

Homework 1

Due September 2, 2025.

Problem 1.

Do exercise 1.3 of the notes.

Problem 2.

Find all symmetries of a square in a 2-dimensional plane (both rotations and (generalized) reflections). Is parity (space inversion) a rotational symmetry or a generalized reflection? Is parity a rotational or reflectional symmetry of a cube? Construct the Latin square for the complete symmetry group of the square (read the first part of section 2.1).

Problem 3.

Show that a finite group of even order always contains a group element $a \neq e$ satisfying $aa = e$. (*Hint:* Pair the group elements a and a^{-1} if $a^{-1} \neq a$.)